Experiment 1: Library Management System

Aim

To design and develop a library website using HTML, CSS and JavaScript that lets the librarian add, update and delete books, issue books, record returns and make fines, and upload year/semester-wise material; and lets students register, log in, search books, view borrow history and fines, search and download material, and view their profile.

Algorithm

1. Start and display the login page with a role selector (Student / Librarian Admin) and a student registration form.
2. Registration: validate that all fields are filled, the email is valid and the username is unique; then store the student.
3. Login: match the username and password against the admin or student list. If no match, show an error; otherwise open the dashboard for that role.
4. Admin - Books: add a book (title, author, copies), edit it, or delete it (refused if a copy is currently issued).
5. Admin - Issue: choose a student and an available book; set issue date = today and expiry date = issue date + 14 days.
6. Admin - Return: set the return date to today. Fine = late days x Rs. 2, where late days = (return date or today) - expiry date; if fine > 0, "Make Fine" marks it imposed.
7. Admin - Material: add or delete material tagged with year and semester; view all material.
8. Student - Search: filter books by title or author; availability = copies - unreturned issues.
9. Student - View borrow history (issue date, expiry date, status) and fine per book with the total.
10. Student - Filter material by year and semester and download it; view and update the profile.
11. Logout returns to the login page. Stop.

Program (HTML, CSS and JavaScript) - library.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1, viewport-fit=cover">
<title>Library Management System</title>
<style>
:root{box-sizing:border-box;padding-top:env(safe-area-inset-top,0px);padding-bottom:env(safe-area-inset-bottom,0px);
--bg:#f5f6fa;--card:#fff;--tx:#1d2433;--mut:#6b7280;--ac:#2f5fd0;--acx:#fff;--bd:#d9dde7;--bad:#c0392b;--ok:#1e8449}
@media (prefers-color-scheme:dark){:root:not([data-theme="light"]){--bg:#14171f;--card:#1e222d;--tx:#e8eaf0;--mut:#9aa3b5;--
    ac:#6d93f0;--acx:#0d1220;--bd:#343a4a;--bad:#ef7b6e;--ok:#58c183}}
html{scroll-padding-top:env(safe-area-inset-top,0px)}
*{box-sizing:border-box}
body{margin:0;background:var(--bg);color:var(--tx);font-family:system-ui,Segoe UI,Arial,sans-serif;line-height:1.45}
header{background:var(--ac);color:var(--acx);padding:12px 16px;display:flex;justify-content:space-between;align-
    items:center;flex-wrap:wrap;gap:8px}
header h1{font-size:1.1rem;margin:0}
main{max-width:960px;margin:16px auto;padding:0 12px}
.card{background:var(--card);border:1px solid var(--bd);border-radius:10px;padding:16px;margin-bottom:14px}
nav{display:flex;gap:6px;flex-wrap:wrap;margin-bottom:14px}
button,.btn{background:var(--ac);color:var(--acx);border:0;border-radius:6px;padding:8px 13px;cursor:pointer;font-size:.9rem}
button.alt{background:transparent;color:var(--tx);border:1px solid var(--bd)}
button.on{background:var(--ac);color:var(--acx)}
button.sm{padding:4px 9px;font-size:.8rem}
input,select{padding:8px;border:1px solid var(--bd);border-radius:6px;background:var(--bg);color:var(--tx);font-size:.9rem;max-
    width:100%}
.row{display:flex;gap:8px;flex-wrap:wrap;margin-bottom:10px;align-items:center}
.tw{overflow-x:auto}
table{border-collapse:collapse;width:100%;font-size:.88rem}
th,td{text-align:left;padding:7px 9px;border-bottom:1px solid var(--bd);white-space:nowrap}
th{color:var(--mut);font-weight:600}
.bad{color:var(--bad);font-weight:600}.ok{color:var(--ok);font-weight:600}.mut{color:var(--mut);font-size:.85rem}
#login{max-width:380px;margin:30px auto}
#login input{width:100%;margin-bottom:8px}
h2{margin:0 0 12px;font-size:1.05rem}
</style>
</head>
<body>
<header><h1>■ Central Library Management System</h1><div id="who"></div></header>
<main id="app"></main>
<script>
const RATE=2, LOAN=14;
const today=()=>new Date().toISOString().slice(0,10);
const addDays=(d,n)=>{const x=new Date(d);x.setDate(x.getDate()+n);return x.toISOString().slice(0,10)};
const days=(a,b)=>Math.round((new Date(b)-new Date(a))/864e5);
const $=id=>document.getElementById(id);
const esc=s=>String(s).replace(/[&<>"]/g,c=>({'&':'&amp;','<':'&lt;','>':'&gt;','"':'&quot;'}[c]));

const D={
 admins:[{u:'admin',p:'admin123'}],
 users:[{u:'student',p:'stud123',name:'Ravi Kumar',email:'ravi@college.edu',year:2,sem:3}],
 books:[
  {id:1,title:'Introduction to Algorithms',author:'T. Cormen',qty:3},
  {id:2,title:'Operating System Concepts',author:'A. Silberschatz',qty:2},
  {id:3,title:'Computer Networks',author:'A. Tanenbaum',qty:4},
  {id:4,title:'Web Technologies',author:'U. Sharma',qty:2}],
 issues:[],
 mats:[
  {id:1,title:'HTML & CSS Notes',year:2,sem:3,text:'HTML structure, tags, CSS selectors and box model.'},
  {id:2,title:'DBMS Question Bank',year:2,sem:4,text:'Unit-wise DBMS questions.'},
  {id:3,title:'Data Structures Notes',year:1,sem:2,text:'Arrays, stacks, queues, linked lists.'}]
};
let role=null, me=null, tab='', nid=100;

const fineOf=i=>Math.max(0,days(i.due,i.ret||today()))*RATE;
const bookName=id=>(D.books.find(b=>b.id==id)||{title:'?'}).title;
const avail=b=>b.qty-D.issues.filter(i=>i.book==b.id&&!i.ret).length;

function msg(t){alert(t)}
function logout(){role=me=null;tab='';render()}

function render(){
 $('who').innerHTML=me?`${esc(me.name||me.u)} (${role}) <button class="alt sm" style="color:inherit;border-color:currentColor"
    onclick="logout()">Logout</button>`:'';
 if(!me){$('app').innerHTML=loginView();return}
 const tabs=role=='admin'?['Books','Issue Book','Returns & Fines','Update Material','View Material']
  :['Search Book','My Borrow History','Check Fine','Search Material','My Profile'];
 if(!tab||!tabs.includes(tab))tab=tabs[0];
 $('app').innerHTML='<nav>'+tabs.map(t=>`<button class="${t==tab?'on':'alt'}"
    onclick="tab='${t}';render()">${t}</button>`).join('')+'</nav><div id="view"></div>';
 const v={'Books':booksView,'Issue Book':issueView,'Returns & Fines':returnsView,'Update Material':matAdminView,
  'View Material':()=>matView(false),'Search Book':searchView,'My Borrow History':historyView,'Check Fine':fineView,
  'Search Material':()=>matView(true),'My Profile':profileView}[tab];
 $('view').innerHTML=v();
 if(tab=='Search Book')results();
}

/* ---------- login / register ---------- */
function loginView(){return `<div id="login" class="card"><h2>Login</h2>
 <select id="r" style="width:100%;margin-bottom:8px"><option value="user">Student</option><option value="admin">Librarian
```

```
    Admin</option></select>
 <input id="lu" placeholder="Username"><input id="lp" type="password" placeholder="Password">
 <button onclick="doLogin()">Login</button>
 <p class="mut">Demo: admin / admin123  |  student / stud123</p>
 <hr style="border:0;border-top:1px solid var(--bd)"><h2>New student? Register</h2>
 <input id="rn" placeholder="Full name"><input id="re" placeholder="Email"><input id="ru" placeholder="Username"><input id="rp"
    type="password" placeholder="Password">
 <div class="row"><input id="ry" type="number" min="1" max="4" placeholder="Year" style="width:48%"><input id="rs" type="number"
     min="1" max="8" placeholder="Semester" style="width:48%"></div>
 <button onclick="doReg()">Register</button></div>`}
function doLogin(){
 const r=$('r').value,u=$('lu').value.trim(),p=$('lp').value;
 const f=(r=='admin'?D.admins:D.users).find(x=>x.u==u&&x.p==p);
 if(!f)return msg('Invalid username or password');
 role=r;me=f;tab='';render();
}
function doReg(){
 const o={name:$('rn').value.trim(),email:$('re').value.trim(),u:$('ru').value.trim(),p:$('rp').value,year:+$('ry').value,sem:+$
    ('rs').value};
 if(!o.name||!o.u||!o.p||!o.year||!o.sem)return msg('Please fill all fields');
 if(!/^\S+@\S+\.\S+$/.test(o.email))return msg('Enter a valid email');
 if(D.users.some(x=>x.u==o.u))return msg('Username already taken');
 D.users.push(o);msg('Registered. You can login now.');
}

/* ---------- admin ---------- */
function booksView(){
 return `<div class="card"><h2>Add / Update Book</h2><div class="row">
 <input id="bid" type="hidden"><input id="bt" placeholder="Title"><input id="ba" placeholder="Author"><input id="bq"
    type="number" min="1" placeholder="Copies" style="width:90px">
 <button onclick="saveBook()">Save</button></div></div>
 <div class="card
    tw"><h2>Catalog</h2><table><tr><th>ID</th><th>Title</th><th>Author</th><th>Copies</th><th>Available</th><th></th></tr>`+
 D.books.map(b=>`<tr><td>${b.id}</td><td>${esc(b.title)}</td><td>${esc(b.author)}</td><td>${b.qty}</td><td>${avail(b)}</td>
 <td><button class="sm alt" onclick="editBook(${b.id})">Edit</button> <button class="sm alt"
    onclick="delBook(${b.id})">Delete</button></td></tr>`).join('')+'</table></div>';
}
function saveBook(){
 const t=$('bt').value.trim(),a=$('ba').value.trim(),q=+$('bq').value,id=$('bid').value;
 if(!t||!a||q<1)return msg('Enter title, author and copies');
 if(id){Object.assign(D.books.find(b=>b.id==id),{title:t,author:a,qty:q})}else D.books.push({id:++nid,title:t,author:a,qty:q});
 render();
}
function editBook(id){const b=D.books.find(x=>x.id==id);$('bid').value=id;$('bt').value=b.title;$('ba').value=b.author;$('bq').v
    alue=b.qty;window.scrollTo(0,0)}
function delBook(id){
 if(D.issues.some(i=>i.book==id&&!i.ret))return msg('Cannot delete: book is currently issued');
 D.books=D.books.filter(b=>b.id!=id);render();
}
function issueView(){
 return `<div class="card"><h2>Issue Book to Student</h2><div class="row">
 <select id="is">${D.users.map(u=>`<option value="${esc(u.u)}">${esc(u.name)} (${esc(u.u)})</option>`).join('')}</select>
 <select id="ib">${D.books.filter(b=>avail(b)>0).map(b=>`<option value="${b.id}">${esc(b.title)} (${avail(b)}
    left)</option>`).join('')}</select>
 <button onclick="issue()">Issue</button></div><p class="mut">Loan period: ${LOAN} days. Fine: Rs. ${RATE} per day after the
    expiry date.</p></div>`;
}
function issue(){
 const b=$('ib').value;if(!b)return msg('No book available');
 D.issues.push({id:++nid,user:$('is').value,book:+b,issued:today(),due:addDays(today(),LOAN),ret:null,fined:false});
 msg('Book issued. Expiry date: '+addDays(today(),LOAN));render();
}
function returnsView(){
 return `<div class="card tw"><h2>Issued Books &amp;amp; Return Status</h2><table><tr><th>Student</th><th>Book</th><th>Issue
    Date</th><th>Exp Date</th><th>Status</th><th>Fine</th><th></th></tr>`+
 (D.issues.length?D.issues.map(i=>{const f=fineOf(i);return
    `<tr><td>${esc(i.user)}</td><td>${esc(bookName(i.book))}</td><td>${i.issued}</td><td>${i.due}</td>
 <td>${i.ret?'<span class="ok">Returned '+i.ret+'</span>':(f?'<span class="bad">Overdue</span>':'Issued')}</td>
 <td>Rs. ${f}${i.fined?' (imposed)':''}</td>
 <td>${i.ret?'':`<button class="sm" onclick="ret(${i.id})">Return</button> `}${f&&!i.fined?`<button class="sm alt"
    onclick="mkFine(${i.id})">Make Fine</button>`:''}</td></tr>`}).join(''):'<tr><td colspan="7" class="mut">No books issued
    yet.</td></tr>')+
 `</table><div class="row" style="margin-top:12px"><span class="mut">Demo helper: back-date an issue to test fines
    →</span><select id="bk">${D.issues.map(i=>`<option value="${i.id}">${esc(i.user)} -
    ${esc(bookName(i.book))}</option>`).join('')}</select><button class="alt sm" onclick="backdate()">Make 10 days
    overdue</button></div></div>`;
}
function ret(id){const i=D.issues.find(x=>x.id==id);i.ret=today();if(fineOf(i))i.fined=true;render()}
function mkFine(id){D.issues.find(x=>x.id==id).fined=true;render()}
function backdate(){const
    i=D.issues.find(x=>x.id==$('bk').value);if(!i)return;i.issued=addDays(today(),-(LOAN+10));i.due=addDays(today(),-
    10);render()}

function matAdminView(){
 return `<div class="card"><h2>Add Material (year / semester wise)</h2><div class="row">
 <input id="mt" placeholder="Title"><input id="my" type="number" min="1" max="4" placeholder="Year" style="width:80px"><input
    id="ms" type="number" min="1" max="8" placeholder="Sem" style="width:80px">
 <input id="mx" placeholder="Content / description"><button onclick="addMat()">Add</button></div></div>`+matView(false);
}
function addMat(){
 const m={id:++nid,title:$('mt').value.trim(),year:+$('my').value,sem:+$('ms').value,text:$('mx').value.trim()};
 if(!m.title||!m.year||!m.sem)return msg('Enter title, year and semester');D.mats.push(m);render();
}
```

```
function delMat(id){D.mats=D.mats.filter(m=>m.id!=id);render()}

/* ---------- material (shared) ---------- */
function matView(filter){
 const rows=D.mats.map(m=>`<tr><td>${esc(m.title)}</td><td>${m.year}</td><td>${m.sem}</td>
 <td>${filter?`<button class="sm" onclick="dl(${m.id})">Download</button>`:`<button class="sm alt"
    onclick="delMat(${m.id})">Delete</button>`}</td></tr>`).join('');
 return `<div class="card tw"><h2>${filter?'Search Material':'Study Material'}</h2>
 ${filter?`<div class="row"><select id="fy" onchange="fm()"><option value="">All
    years</option>${[1,2,3,4].map(y=>`<option>${y}</option>`)}</select>
 <select id="fs" onchange="fm()"><option value="">All
    semesters</option>${[1,2,3,4,5,6,7,8].map(y=>`<option>${y}</option>`)}</select></div>`:''}
 <table id="mtab"><tr><th>Title</th><th>Year</th><th>Semester</th><th></th></tr>${rows}</table></div>`;
}
function fm(){
 const y=$('fy').value,s=$('fs').value;
 $('mtab').innerHTML='<tr><th>Title</th><th>Year</th><th>Semester</th><th></th></tr>'+
 D.mats.filter(m=>(!y||m.year==y)&&(!s||m.sem==s)).map(m=>`<tr><td>${esc(m.title)}</td><td>${m.year}</td><td>${m.sem}</td><td><b
    utton class="sm" onclick="dl(${m.id})">Download</button></td></tr>`).join('');
}
function dl(id){
 const m=D.mats.find(x=>x.id==id);
 const a=document.createElement('a');
 a.href=URL.createObjectURL(new Blob([m.title+'\nYear '+m.year+', Semester '+m.sem+'\n\n'+m.text],{type:'text/plain'}));
 a.download=m.title.replace(/\W+/g,'_')+'.txt';a.click();
}

/* ---------- student ---------- */
function searchView(){
 return `<div class="card"><h2>Search Book</h2><div class="row"><input id="q" placeholder="Title or author" oninput="results()"
    style="flex:1"></div>
 <div class="tw" id="res"></div></div>`;
}
function results(){
 const q=($('q')||{value:''}).value.toLowerCase();
 const l=D.books.filter(b=>(b.title+' '+b.author).toLowerCase().includes(q));
 $('res').innerHTML='<table><tr><th>Title</th><th>Author</th><th>Status</th></tr>'+
 l.map(b=>`<tr><td>${esc(b.title)}</td><td>${esc(b.author)}</td><td>${avail(b)>0?`<span class="ok">Available
    (${avail(b)})</span>`:'<span class="bad">Not available</span>'}</td></tr>`).join('')+
 (l.length?'':'<tr><td colspan="3" class="mut">No books found.</td></tr>')+'</table>';
}
const mine=()=>D.issues.filter(i=>i.user==me.u);
function historyView(){
 return `<div class="card tw"><h2>My Borrow History</h2><table><tr><th>Book</th><th>Issue Date</th><th>Exp
    Date</th><th>Status</th></tr>`+
 (mine().map(i=>`<tr><td>${esc(bookName(i.book))}</td><td>${i.issued}</td><td>${i.due}</td><td>${i.ret?'Returned '+i.ret:'Not
    returned'}</td></tr>`).join('')||'<tr><td colspan="4" class="mut">No borrow history.</td></tr>')+'</table></div>';
}
function fineView(){
 const l=mine().filter(i=>fineOf(i)>0),t=l.reduce((s,i)=>s+fineOf(i),0);
 return `<div class="card tw"><h2>Check Fine</h2><table><tr><th>Book</th><th>Exp Date</th><th>Days Late</th><th>Fine</th></tr>`+
 (l.map(i=>`<tr><td>${esc(bookName(i.book))}</td><td>${i.due}</td><td>${fineOf(i)/RATE}</td><td>Rs.
    ${fineOf(i)}</td></tr>`).join('')||'<tr><td colspan="4" class="ok">No fine. </td></tr>')+
 `</table><p><b>Total fine: <span class="${t?'bad':'ok'}">Rs. ${t}</span></b></p></div>`;
}
function profileView(){
 return `<div class="card"><h2>My Profile</h2><div class="row"><input id="pn" value="${esc(me.name)}"><input id="pe"
    value="${esc(me.email)}">
 <input id="py" type="number" value="${me.year}" style="width:80px"><input id="ps" type="number" value="${me.sem}"
    style="width:80px"><button onclick="saveProf()">Update</button></div>
 <p class="mut">Username: ${esc(me.u)}</p></div>`;
}
function saveProf(){me.name=$('pn').value;me.email=$('pe').value;me.year=+$('py').value;me.sem=+$('ps').value;msg('Profile
    updated');render()}

render();
</script>
</body>
</html>
```

Output

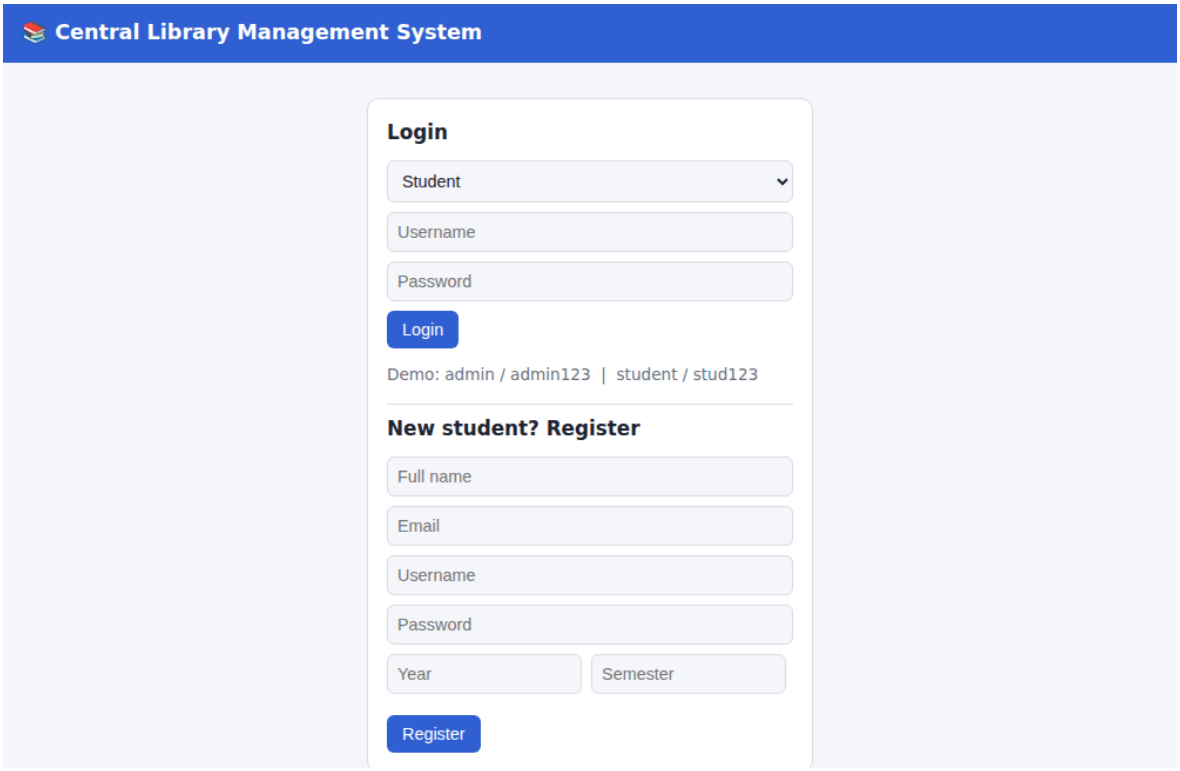


Fig 1: Login and registration page

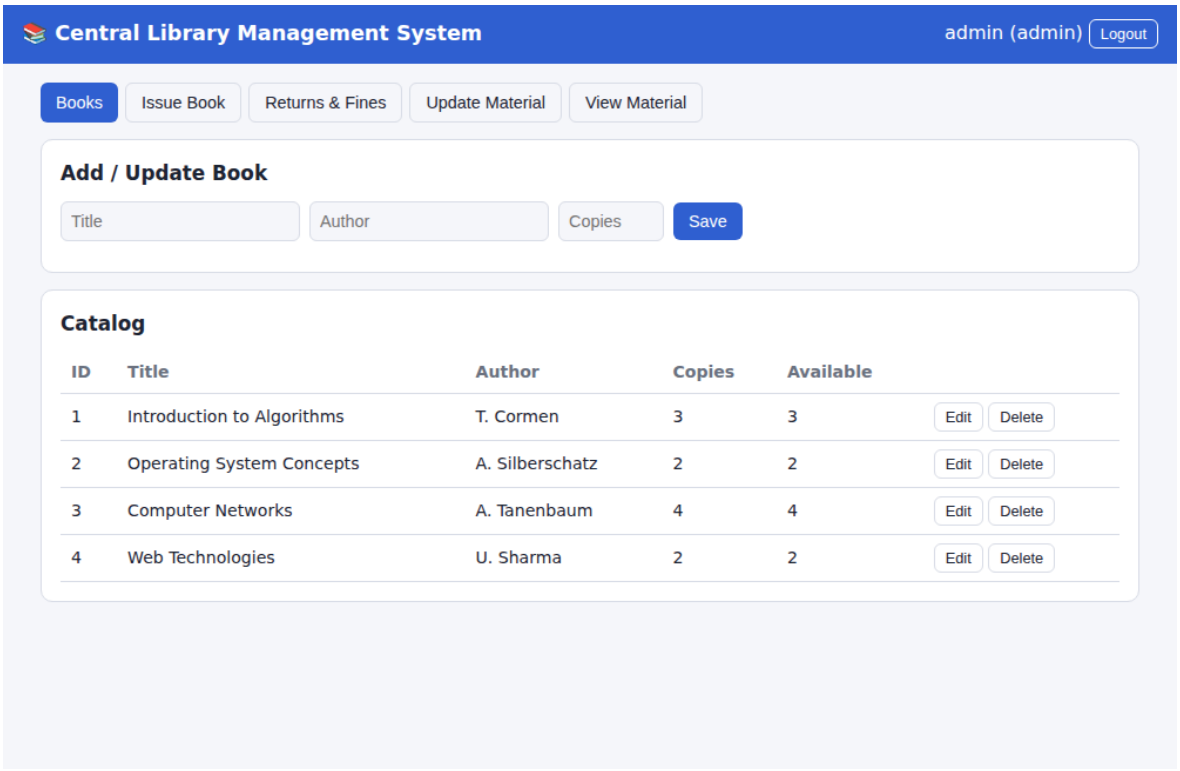


Fig 2: Admin dashboard - Books catalog

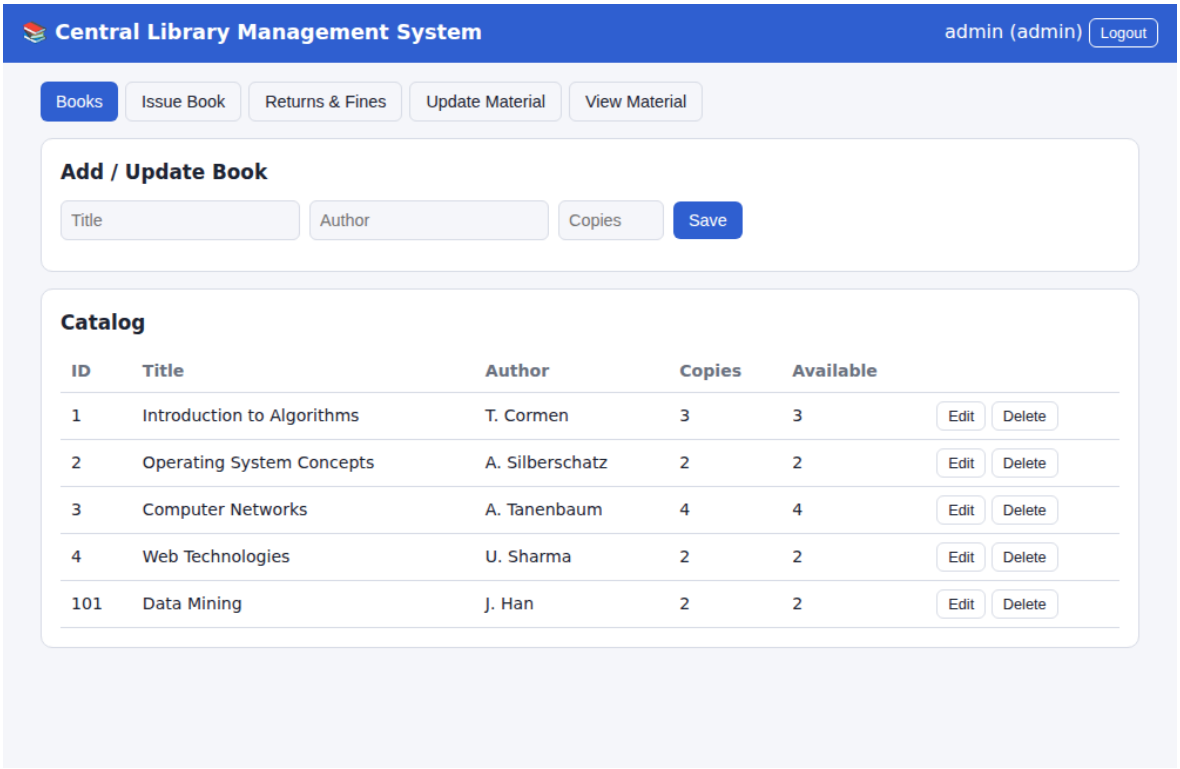


Fig 3: Admin adds a new book (Data Mining)

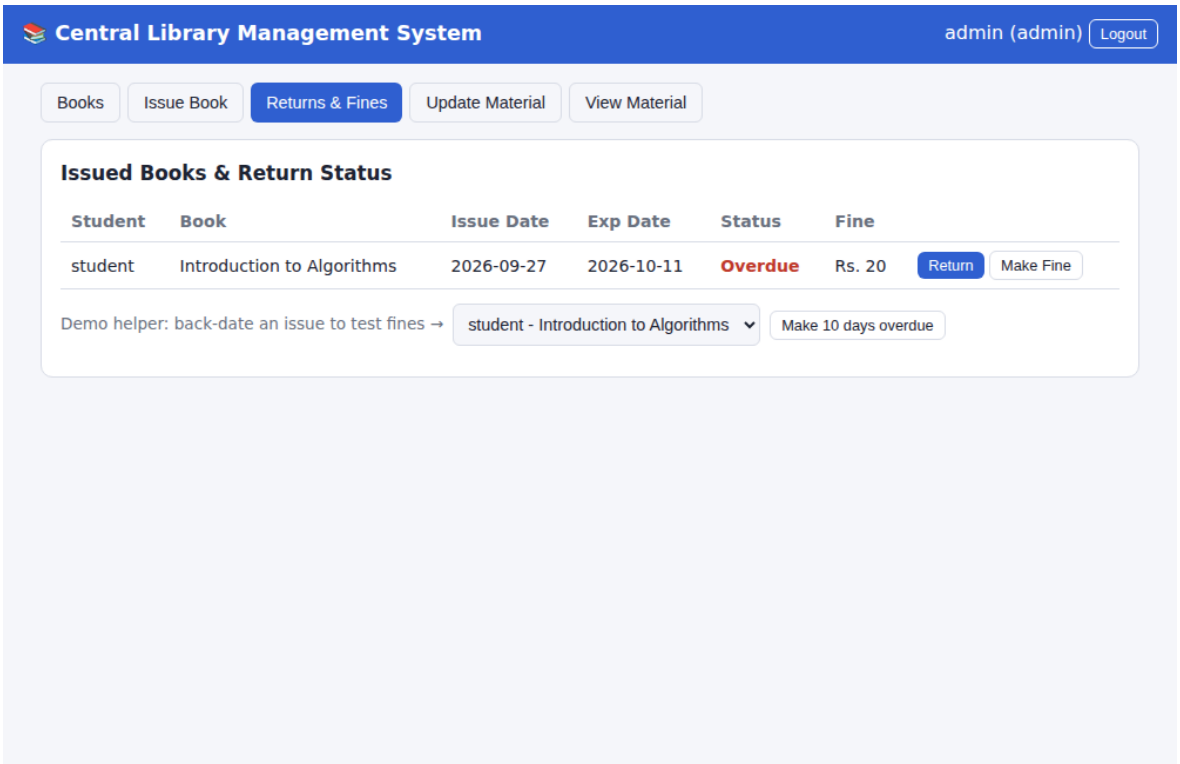


Fig 4: Admin - Issued books, return status and fine (10 days overdue)

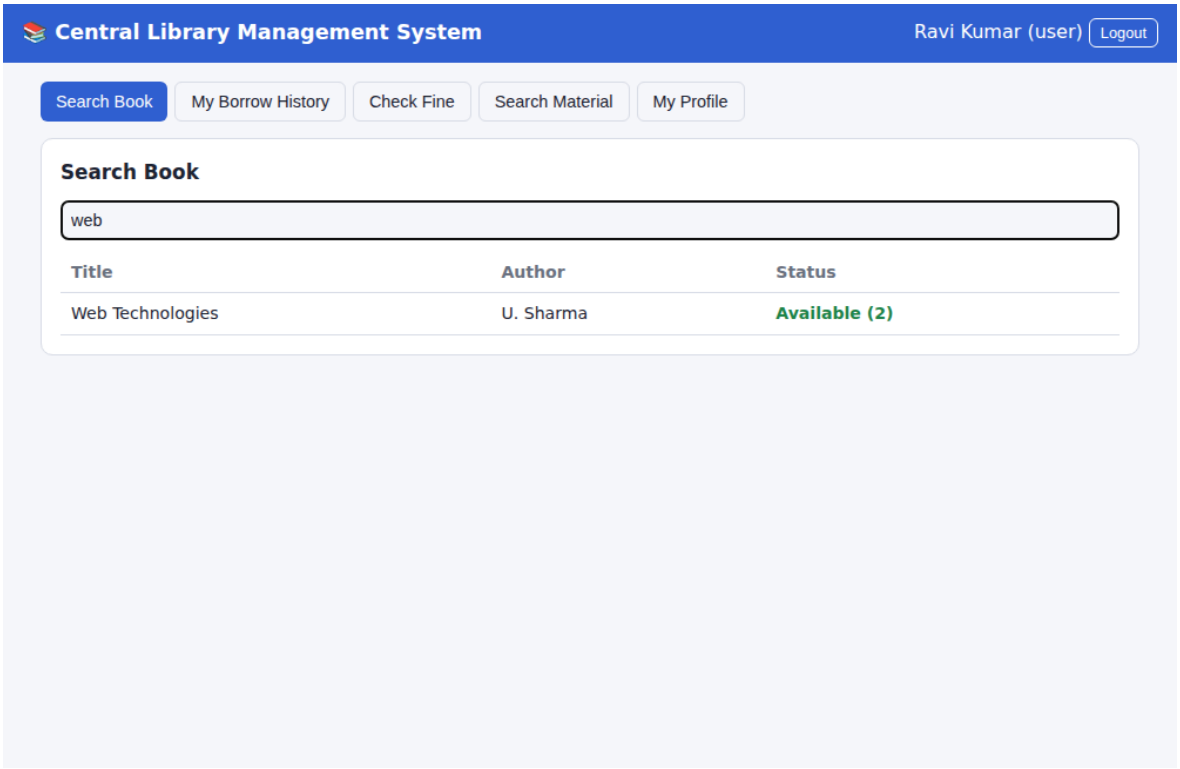


Fig 5: Student - Search Book (query "web")

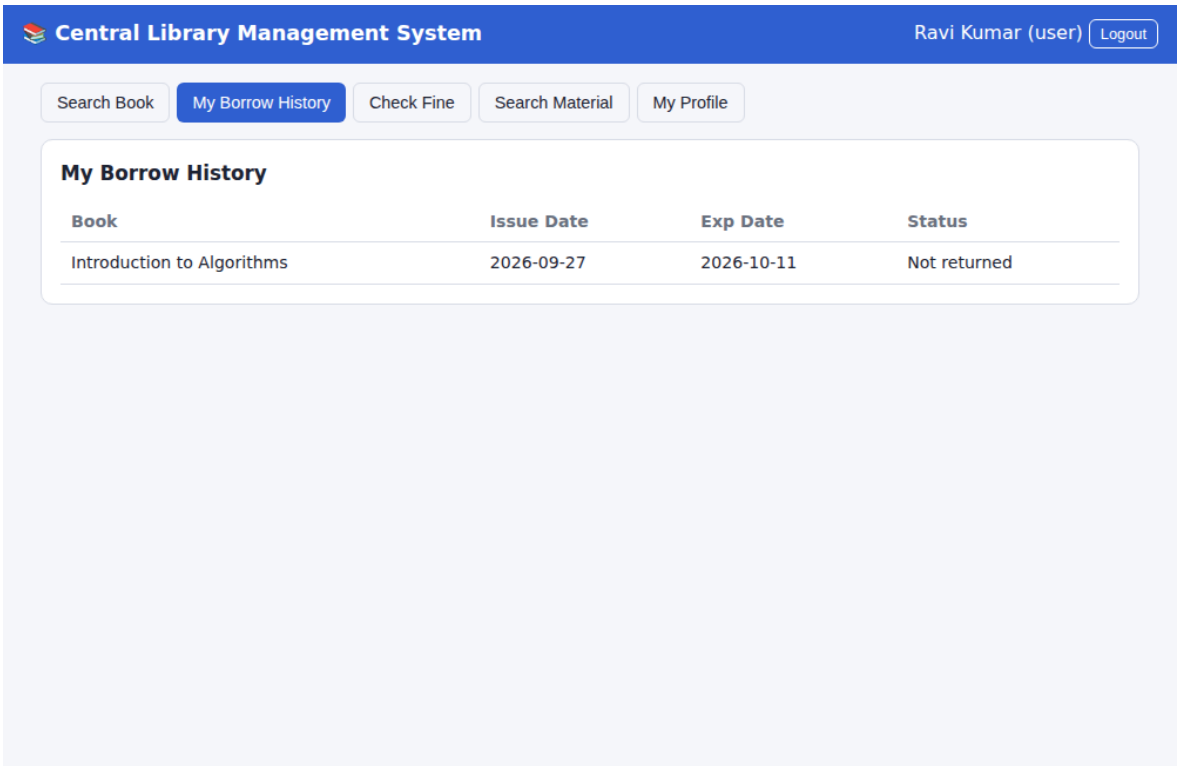


Fig 6: Student - My Borrow History (issue date, expiry date)

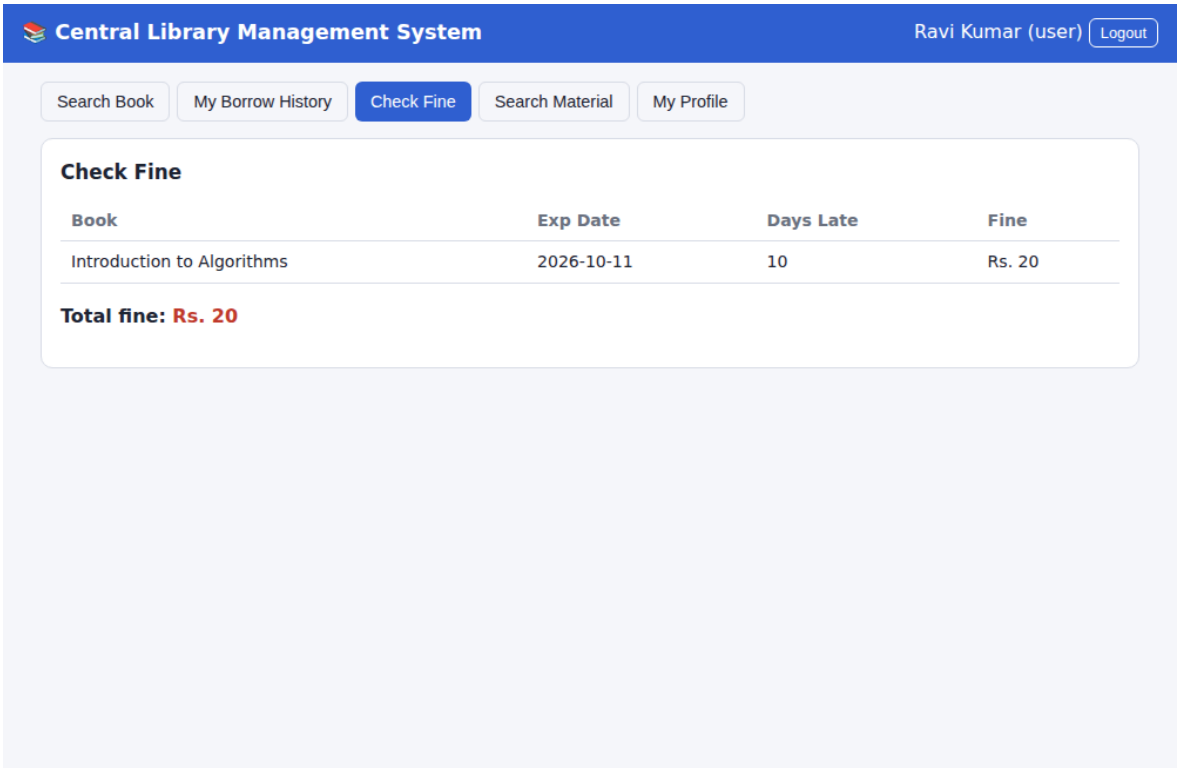


Fig 7: Student - Check Fine (Rs. 20 for 10 days late)

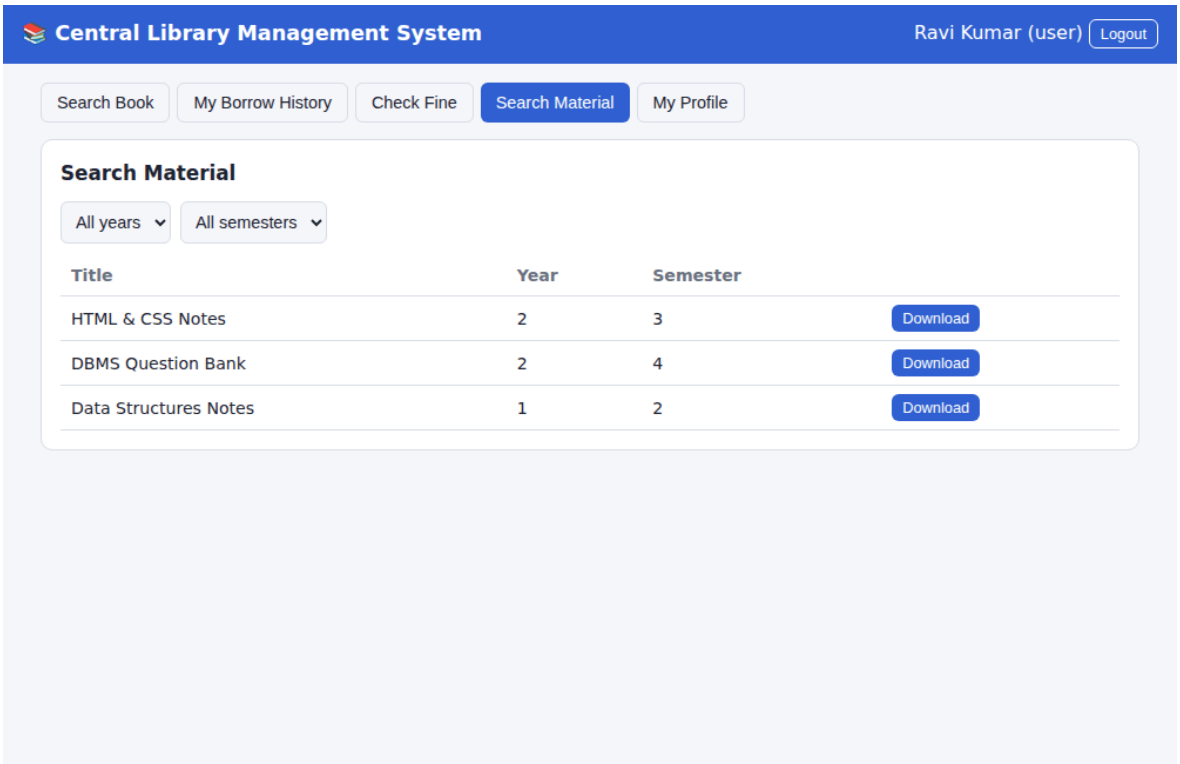


Fig 8: Student - Search Material (year / semester wise)

Result

The library website was developed using HTML, CSS and JavaScript, and the admin operations (add/update/delete book, issue, return, fine, material) and student operations (register, login, search, borrow history, fine, material download, profile) were executed successfully.